

Status_Summary

Helix OTA — Feature Inventory — Status Summary

Revision: 13 **Last modified:** 2026-06-23T09:15:00Z **Companion of:** [Status.md](#)
(Section 11.4.56 two-audience parity).

Video-evidence reconciliation (2026-06-22, §11.4.6 no-bluff). Earlier revisions cited video confirmations at the ephemeral \$HOME/Downloads recording path, which is the §11.4.128 gitignored raw corpus and has been cleared. Durable, committed evidence now is: (1) 10 vision-verified server/e2e control-plane MP4s under docs/qa/20260622-server-recordings-regen/mp4/ (REPORT.md), and (2) the re-proven GREEN QEMU A/B firmware console evidence under docs/qa/20260622T07* (slot-switch, rollback, RAUC dm-verity). On-target RK3588 / Android-agent / real Android A/B rows remain hardware/operator-gated (OTA-004, F55, F56) and are honestly NOT recorded on this host.

Page 1 — For the operator (plain language)

Helix OTA is an enterprise over-the-air update system. Here is what exists and what state it is in.

What works today:

- The **control-plane server** (Go/Gin) is fully built and tested — all APIs for managing devices, releases, deployments, rollouts, recalls, and audit logs have both unit tests and end-to-end tests running in containers. New endpoints: GET /api/v1/devices (device inventory listing) and GET /devices/by-hardware/:hardwareId (hardware-ID reverse lookup) both PASS.
- **Six reusable Go libraries** (protocol types, artifact validation, rollout engine, telemetry, HTTP/3) are built and tested with unit + stress + chaos tests.
- **Container infrastructure** is in place via the vasic-digital/containers submodule — tests run in podman pods.
- **The emulator ladder proves the A/B update core on this Mac without needing the real hardware:**
 - The container round-trip (control plane talking to a virtual device) works.
 - The emulated device boots to a live Linux userspace — full boot transcript captured as proof.

- **The A/B slot switch is proven** — a real U-Boot bootloader switches between system slot A and slot B on demand, with 3/3 identical passes.
- **Automatic rollback on a corrupt slot is proven** — when we mark a slot bad, the bootloader detects it and falls back to the good slot automatically.
- **RAUC dm-verity slot integrity (PWU-AB-2) is proven** — direct-dd A/B slot switch GREEN 3/3 deterministic with captured evidence.
- **ApplyPort (PWU-AB-4) is implemented** — slot manager, ed25519 signature verifier, health marker, HTTP client, and CLI binary — 36 tests, 3 Go source files, 2 Kotlin files, all tested and passing.
- **HelixTrack integration (F108) has begun** — the multi-platform project management system is being integrated as the workable-items engine. Phase 0 complete: launcher scripts, helix-deps.yaml, space config. Three parallel research agents working on Phases 1-3 (multi-space data isolation, containers compose, docs_chain sync). DESIGN status.
- **Security tests, e2e tests, and pre-build gates** all pass.
- **Production deployment** is live on nezha.local — a full 3-container stack (server, PostgreSQL, SPA) orchestrated via docker-compose.
- **Remote stress testing** proves 291 req/s sustained device registration with 100/100 devices and all stress/chaos tests passing.
- **Security hardening** — IDOR project-scoped authorization prevents cross-project access; Tauri IPC scoped to minimum permissions; docker-compose secrets no longer ship default credentials.
- **MountManagerUI bug fixed** — the SPA now serves correctly at /manager/.
- **Real RK3588 hardware talked to the control plane (F113)** — a physical RK3588 Android-15 board (over Ethernet) registered itself, checked for an update, and sent telemetry to the server, and the server confirms the board's own telemetry updated its state (update_state=success). Proof: docs/qa/20260622-rk3588-controlplane/REPORT.md. The second board (Wi-Fi) was honestly skipped — its network path is blocked by a VPN/AP-isolation issue, not a Helix problem. Important honesty note: **these boards are single-slot, so this proves the control plane on real hardware, NOT the actual A/B firmware swap** — that is the Cuttlefish path's job (below). No changes were made to the boards.
- **Commit/push cascade fixed (F117)** — the commit wrapper now reliably fans out every commit to all four upstream mirrors with an honest exit on failure. This is genuinely real: multiple real commits were pushed to all four mirrors this session through the fix.
- **Governance** (Issues, Fixed, CONTINUATION, README, Status docs) is maintained and in sync.

What is still pending (NOT done yet — honestly):

- **Android OTA agent on-device verification** — the two Kotlin modules exist and the ApplyPort Go/Kotlin code is implemented but has not been tested against a real Android target.
- **Cuttlefish Tier-2 Android A/B (F112 / F55) — DONE, now VERIFIED.** As of 2026-06-23 the real end-to-end Android A/B update ran on a live Cuttlefish virtual device on the nezha host: a real ~1 GB update was applied, the device switched to the new slot (_a→_b), and when the old slot was deliberately broken the device safely fell back to the known-good slot — exactly how a real phone protects itself from a bad update. The update file was downloaded with no passwords needed. This is the deepest test we can run without a physical board. Rootless containers cannot host Cuttlefish, so a narrow rootful-privileged exception is documented (docs/design/CUTTLEFISH_ROOTFUL_EXCEPTION.md). **The bring-up is now runbook-ready**

— docs/design/CUTTLEFISH_NEZHA_RUNBOOK.md gives the exact step-by-step for the real nezha run: the operator runs the three privileged steps (nezha has no passwordless sudo), and the agent drives the rest (extract assets, launch the virtual device, run the A/B + auto-rollback check). It stays NOT-a-real-A/B-pass until that runbook is actually executed with the slot-flip + rollback evidence captured.

- **Container stack distribution (F114) — REAL deploy to thinker is GREEN (VERIFIED).** A fully-automated, non-dry-run `HELIXTRACK_REMOTE_HOST=thinker.local bash scripts/distribute_stack.sh` ran end-to-end: it **built** the helixtrack-core image on thinker (rootless podman-compose) from the Go 1.24 Dockerfile, brought the stack up, and a fresh container reported `podman ps: helixtrack-core Up (healthy) + helixtrack-postgres Up (healthy)`, with `curl -sf http://localhost:8080/health → HTTP 200 {"status":"ok"} (FailingStreak=0)`. Evidence docs/qa/20260622-222645-distribute-thinker-FULLY-GREEN/. The fix chain that made it green: `distribute_stack.sh` (provider-preference for podman-compose + nested-mkdir + build-before-up + down-before-up idempotency); the containers submodule healthcheck on /health (dcef56d); and the HelixTrack Core Dockerfile bumped to `golang:1.24`, its gutted source restored (3c62217/3483699), with `curl` added to the runtime image (d0f4bfb) — that closed the prior Go-version blocker. `thinker.local` is the proven live rootless-podman target; the amber docker-fallback path (F115) has not yet been run.
- **Docker fallback for amber (F115) — gate logic verified, real deploy NOT run.** amber has docker but no rootless podman, so it gets an **operator-authorized docker fallback** (explicit opt-in `HELIX_ALLOW_DOCKER_FALLBACK=1`, never the default — rootless-podman is always preferred). The default-OFF gate + host selection are verified in **dry-run** (amber selected via docker only with the flag, honestly skipped without it), but **no real docker deploy to amber has executed**. amber was onboarded (SSH key + docker present) on 2026-06-22.
- **Remote emulator launch hardening (F116) — source done, persistence NOT verified.** The Android emulator launch on nezha was hardened with `setsid` full SSH-session detachment (applied + code-reviewed), and the earlier AVD boot was proven. BUT the actual on-target behaviour the fix targets — the emulator surviving the launching SSH session ending — is **operator-attended and NOT yet run-verified**.
- **Full Android OTA test (Tier-2 Cuttlefish) — cannot run on this Mac** (needs Linux with KVM). Ready for the operator's Linux machine.
- **Real hardware testing (Tier-3 RK3588 board) — no board on the bench yet.**
- **Stress/chaos coverage — present for some submodules but not yet for the server endpoints.**
- **Build resource statistics tracking — mandated but not yet built.**
- **Workable-items SQLite database — mandated but not yet built.**
- **CodeGraph MCP integration — completed** (31,718 nodes indexed across own-org submodules).

Bottom line: The server, Go libraries, remote deployment, remote stress testing, the A/B update core (slot switch + auto-rollback + RAUC dm-verity), and ApplyPort server-side implementation are all proven with captured evidence or testing. Security hardening (IDOR, Tauri IPC, docker-compose secrets) is complete. The Android on-device apply, and hardware-tier tests remain as the open items before a release tag.

Page 2 — For software engineers

Feature inventory summary (all 109 items from Status.md):

Status	Count	Key Items
PASS	50	All server handlers (F01-F34), emulator Tier-0/ Tier-1 (F44-F49), e2e tests (F57-F67), build gates (F69-F71), scripts (F74-F76), Multi-Project API + IDOR (F90-F91), MountManagerUI (F98), IDOR Security (F99), Tauri IPC (F100), Docker Secrets (F101), Remote Deploy (F103), Devices List API (F104), Hardware ID Reverse Lookup (F105), RK3588 control-plane validation — Device B (F113)
SKIP	1	Demo Re-recordings (F107) — stale/rotated per §11.4.154 Go submodules (F35-F41), Tier-2 Cuttlefish driver (F55 — REAL Android A/B on a live cvd, nezha 2026-06-23) , containers (F68), .gitignore (F73), governance (F77-F85), CodeGraph wired (F88), frontend build + tests (F92-F93), Production Deploy (F96), Remote Stress (F97), §11.4.159 Recording Compliance (F106), HelixTrack Integration (F108), Build Resource Stats (F109), Cuttlefish pkg/ cuttlefish (F112 — REAL A/B PASS on a live cvd, nezha 2026-06-23) , Container stack distribution (F114 — REAL fully-automated deploy to thinker GREEN: (heal thy) + /health 200), commit_all cascade-
VERIFIED	27	

Status	Count	Key Items
PROVEN	6	push (F117 — real commits pushed to all four mirrors this session) PWU-AB-1 base+boot (F50), slot switch (F51), auto-rollback (F52), PWU-AB-2 RAUC dm-verity (F53), slot switch video (F94), rollback video (F95)
IMPLEMENTED	2	PWU-AB-4 ApplyPort (F54), ApplyPort Scaffold (F102)
DESIGN	2	ota-android-agent (F42), ota-update-engine-bridge (F43)
OPERATOR-BLOCKED	1	Tier-3 HW (F56 — needs board)
PARTIAL	3	Stress+chaos coverage (F86 — partial submodule set), Docker-fallback distribution §11.4.161 (F115 — gate logic dry-run-verified, real docker deploy to amber NOT run), Remote-emulator full-detachment §11.4.144 (F116 — setsid source hardening applied + reviewed, on-target persistence NOT run-proven)
NOT_STARTED	2	Build-resource-stats (F72), workable-items DB (F87)

F113 (RK3588 control-plane validation) — PASS, honest boundary. Device B (Ethernet, serial 1acdceab90248933) on real RK3588 Android-15 originated GET /healthz 200, GET /api/v1/client/update 204 (no active deployment — correct), POST /api/v1/client/telemetry 202 against a **rootless** linux/amd64 ota-server on nezha, with sink-side GET /devices/by-hardware/... returning update_state=success (board's own POST mutated server state). Device A (Wi-Fi, 19bbb528a1dbbc4d) is an honest topology **SKIP** (VPN tun1 full-tunnel / Wi-Fi AP isolation — busybox nc to both :18080 and :22 time out → blocked path, captured root cause, NOT a Helix defect). **Both boards are NON-A/B** (single-slot, no update_engine) — this validates the control plane on real hardware, NOT native A/B apply (F112/F55's job — now **VERIFIED on nezha 2026-06-23**). ZERO device state changes (§11.4.122/§11.4.133). Evidence: docs/qa/20260622-rk3588-controlplane/REPORT.md.

Distribution mechanism (F114 VERIFIED / F115 PARTIAL) (§11.4.5/§11.4.69). `distribute_stack.sh` (helix layer, §11.4.28) deploys the HelixTrack stack over SSH + remote rootless `podman-compose`. **F114 is now VERIFIED on a REAL fully-automated GREEN deploy to thinker.local** (evidence docs/qa/20260622-222645-distribute-thinker-FULLY-GREEN/): a non-dry-run `HELIXTRACK_REMOTE_HOST=thinker.local bash scripts/distribute_stack.sh` **built** the `helixtrack-core` image on `thinker` (rootless `podman-compose`) from the Go 1.24 Dockerfile, brought the stack up, and a fresh container reported `podman ps: helixtrack-core Up (healthy) + helixtrack-postgres Up (healthy)`, with `curl -sf http://localhost:8080/health → HTTP 200 {"status":"ok"}` (FailingStreak=0; `podman_ps.txt` + `health_body.txt`). **Fix chain:** `distribute_stack.sh` (provider-preference for `podman-compose` + `nested-mkdir` + `build-before-up` + `down-before-up` idempotency); containers-submodule healthcheck on `/health` (dcef56d); HelixTrack Core Dockerfile `go lang:1.24` + restored gutted source (3c62217/3483699) + `curl` in the runtime image (d0f4bfb) — that closed the Rev-19 Go-version blocker. Honest §11.4.6/§11.4.28: the `deploy.log/deploy_tail.txt` from the idempotent re-run show the script's own 60s health-probe printing a timeout line during the down/up churn; the authoritative fresh-container `podman_ps.txt+health_body.txt` (after settle) show (healthy)+200. The helix-layer `compose` file living in the containers submodule (dcef56d) is an operator-approved documented §11.4.28 exception. **F115 stays PARTIAL:** the §11.4.161 operator-authorized docker fallback (`HELIX_ALLOW_DOCKER_FALLBACK=1`, default-OFF rootless-or-nothing, §11.4.112 documented constraint = no rootless `podman` on amber yet) has its **gate logic + host-selection dry-run-verified only** — no real docker deploy to amber has run. `thinker.local` is the proven LIVE rootless-`podman` target; `amber.local` onboarded 2026-06-22 (SSH key + docker); `nezha.local` is read/import-only (NOT a distribution target).

Infra fixes (F116/F117). F116 (PARTIAL): the remote AVD launch on `nezha` was wrapped in `setsid` for full SSH-session detachment (§11.4.144) and **code-reviewed** — boot PROVEN (qa-results/20260622T071848Z-nezha-android-ab/), but on-target **persistence-after-session-end verification is operator-attended / NOT run-proven**, so the runtime persistence claim is not asserted here; only the source-level detachment hardening is done. F117 (VERIFIED — genuinely real): `commit_all.sh/push_all.sh` cascade-push fixed (portable `mkdir` lock + honest exit + per-remote fetch timeout; four-upstream fan-out per §2.1/§11.4.88) — multiple **real commits** (e.g. 17cbd47a, 74af4684) pushed to all four mirrors this session through the fix.

Cuttlefish Tier-2 REAL Android A/B — VERIFIED on nezha 2026-06-23 (F112 / F55). The runbook docs/design/CUTTLEFISH_NEZHA_RUNBOOK.md was executed: the operator ran the privileged container launch (`cf-launch.sh`, no passwordless `sudo` on `nezha`) and the agent drove the A/B flow over `adb -s 127.0.0.1:6520`. A real ~1 GB OTA payload (no-creds androidbuildinternal pre-signed GCS URL, size+md5 verified) was applied through `update_engine` to a live `cvd` (build 15660610, Virtual A/B + verity enforcing, 15 A/B partitions): `onPayloadApplicationComplete(kSuccess) → UPDATED_NEED_REBOOT → slot flip _a→_b` (VAB merge merging→none, `_b` successful) → forced-bad slot `_a` (`bootctl set-slot-as-unbootable` + bounded 256 KB inactive-slot write, §11.4.133) rejected → device booted known-good `_b`. **F112 PARTIAL→VERIFIED; F55 OPERATOR-BLOCKED→VERIFIED.** Evidence docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md (read-the-screen verified, §11.4.158); §11.4.135 guard tests/regression/guard_cuttlefish_ab_proven.sh GREEN (§11.4.107/§11.4.108/§11.4.69). `cvd` left running on `nezha`.

Proven A/B core (captured evidence): - **PWU-AB-1 slot switch** — docs/qa/20260611T094958Z-ab-slot-switch/ (3/3 deterministic PASS, real U-Boot 2024.01 on QEMU virt + HVF) + MP4 recording (1.3 MB) - **PWU-AB-3 auto-rollback** — docs/qa/20260611T095918Z-ab-rollback/ (bad slot -> bootcount exceeded -> altbootcmd swap -> known-good slot; CONTROL proves rollback only fires on bad slot) + MP4 recording (1.4 MB) - **PWU-AB-2 RAUC dm-verity** — docs/qa/20260620T051026Z-ab-rauc-verity/ (direct-dd A/B slot switch, 3/3 deterministic, dd clone rc=0, fw_setenv rc=0, post-slot HELIX_POSTSLOT=B confirmed) - **Video recordings** — All content-verified per §11.4.158 liveness battery.

Pending (honest per Section 11.4.6): - PWU-AB-2 RAUC dm-verity: **PROVEN** — **GREEN 3/3 deterministic** via direct-dd - PWU-AB-4 ApplyPort: **IMPLEMENTED** — slot manager, signature verifier, health marker, HTTP client, CLI binary — 36 tests, 3 Go + 2 Kotlin files; NOT yet tested against a real Android target - Tier-2: tier2_cuttlefish_ab.sh authored, SKIPS on this host (no /dev/kvm) - Tier-3: No physical board

No release tag — Section 11.4.40 requires full ladder GREEN; Tier-2 + Tier-3 remain blocked.

Video Recording Gaps

Priority	Gap	Depends On
High	Tier-2 Cuttlefish Android OTA apply screen recording	nezha Linux+KVM — RUNBOOK-READY (docs/design/CUTTLEFISH_NEZHA_RUNBOOK.md); operator runs 3 privileged steps, agent drives the rest
High	Tier-3 RK3588 HDMI capture of on-device OTA	Physical board
Medium	Tier-1 AVD + HVF screen recording	Existing qa evidence may be partial
Low	Tier-0 container round-trip	Console logs suffice
Fixed	A/B slot switch + rollback	2 MP4 recordings captured (1.3 MB + 1.4 MB), content-verified per §11.4.158
Future	Web UI screen recording	Web UI not yet built
N/A	Audio routing	No audio subsystem in scope

Section-refs

Section 11.4.45 (Status doc), Section 11.4.56 (two-audience summary), Section 11.4.5 (captured-evidence table), Section 11.4.6 (no-guessing), Section 11.4.44 (revision header), Section 11.4.65 (universal export).